25.1 Introduction

The Motor Control PWM (MCPWM) is optimized for three-phase AC and DC motor control applications, but can be used in many other applications that need timing, counting, capture, and comparison.

25.2 Description

The MCPWM contains three independent channels, each including:

- a 32-bit Timer/Counter (TC)
- a 32-bit Limit register (LIM)
- a 32-bit Match register (MAT)
- a 10-bit dead-time register (DT) and an associated 10-bit dead-time counter
- a 32-bit capture register (CAP)
- two modulated outputs (MCOA and MCOB) with opposite polarities
- a period interrupt, a pulse-width interrupt, and a capture interrupt

Input pins MCI0-2 can trigger TC capture or increment a channel's TC. A global Abort input can force all of the channels into "A passive" state and cause an interrupt.

25.3 Pin description

Table 454 lists the MCPWM pins.

Table 454. Pin summary

Pin	Type	Description
MCOA0, MCOA1, MCOA2	O	Output A for channels 0, 1, 2
MCOB0, MCOB1, MCOB2	O	Output B for channels 0, 1, 2
$\overline{\text{MCABORT}}$	I	Low-active Fast Abort
MCI0, MCI1, MCI2	I	Input for channels 0, 1, 2

25.4 Block Diagram

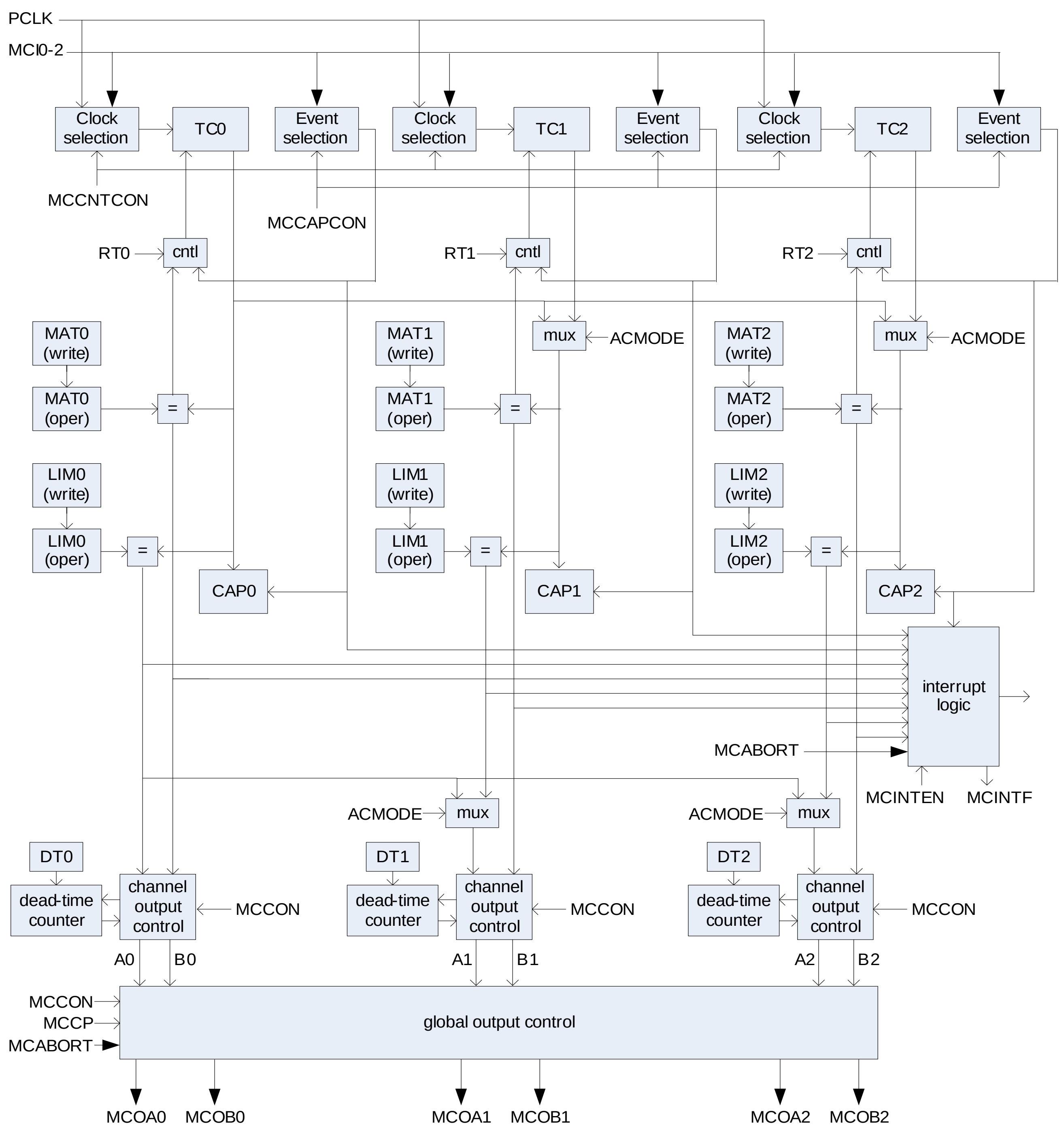

Fig 122. MCPWM Block Diagram

25.5 Configuring other modules for MCPWM use

Configure the following registers in other modules before using the Motor Control PWM:

1. Power: in the PCONP register (Table 46), set bit PCMCPWM.
   Remark: On reset the MCPWM is disabled (PCMCPWM = 0).
2. Peripheral clock: in the PCLKSEL1 register (Table 40) select PCLK_MCPWM.
3. Pins: select MCPWM functions through the PINSEL registers. Select modes for these pins through the PINMODE registers (Section 8.5).
4. Interrupts: See Section 25.7.3 for motor control PWM related interrupts. Interrupts can be enabled in the NVIC using the appropriate Interrupt Set Enable register.

25.6 General Operation

Section 25.8 includes detailed descriptions of the various modes of MCPWM operation, but a quick preview here will provide background for the register descriptions below.

The MCPWM includes 3 channels, each of which controls a pair of outputs that in turn can control something off-chip, like one set of coils in a motor. Each channel includes a Timer/Counter (TC) register that is incremented by a processor clock (timer mode) or by an input pin (counter mode).

Each channel has a Limit register that is compared to the TC value, and when a match occurs the TC is “recycled” in one of two ways. In “edge-aligned mode” the TC is reset to 0, while in “centered mode” a match switches the TC into a state in which it decrements on each processor clock or input pin transition until it reaches 0, at which time it starts counting up again.

Each channel also includes a Match register that holds a smaller value than the Limit register. In edge-aligned mode the channel’s outputs are switched whenever the TC matches either the Match or Limit register, while in center-aligned mode they are switched only when it matches the Match register.

So the Limit register controls the period of the outputs, while the Match register controls how much of each period the outputs spend in each state. Having a small value in the Limit register minimizes “ripple” if the output is integrated into a voltage, and allows the MCPWM to control devices that operate at high speed.

The “downside” of small values in the Limit register is that they reduce the resolution of the duty cycle controlled by the Match register. If you have 8 in the Limit register, the Match register can only select the duty cycle among 0%, 12.5%, 25%, …, 87.5%, or 100%. In general, the resolution of each step in the Match value is 1 divided by the Limit value.

This trade-off between resolution and period/frequency is inherent in the design of pulse width modulators.

25.7 Register description

“Control” registers and “interrupt” registers have separate read, set, and clear addresses. Reading such a register’s read address(e.g. MCCON) yields the state of the register bits. Writing ones to the set address (e.g. MCCON_SET) sets register bit(s), and writing ones to the clear address (e.g. MCCON_CLR) clears register bit(s).

The Capture registers (MCCAP) are read-only, and the write-only MCCAP_CLR address can be used to clear one or more of them. All the other MCPWM registers (MCTIM, MCPER, MCPW, MCDEADTIME, and MCCP) are normal read-write registers.

Table 455. Motor Control Pulse Width Modulator (MCPWM) register map

Name	Description	Access	Reset value	Address
MCCON	PWM Control read address	RO	0	0x400B 8000
MCCON_SET	PWM Control set address	WO	-	0x400B 8004
MCCON_CLR	PWM Control clear address	WO	-	0x400B 8008
MCCAPCON	Capture Control read address	RO	0	0x400B 800C
MCCAPCON_SET	Capture Control set address	WO	-	0x400B 8010
MCCAPCON_CLR	Event Control clear address	WO	-	0x400B 8014
MCTC0	Timer Counter register, channel 0	R/W	0	0x400B 8018
MCTC1	Timer Counter register, channel 1	R/W	0	0x400B 801C
MCTC2	Timer Counter register, channel 2	R/W	0	0x400B 8020
MCLIM0	Limit register, channel 0	R/W	0xFFFF FFFF	0x400B 8024
MCLIM1	Limit register, channel 1	R/W	0xFFFF FFFF	0x400B 8028
MCLIM2	Limit register, channel 2	R/W	0xFFFF FFFF	0x400B 802C
MCMAT0	Match register, channel 0	R/W	0xFFFF FFFF	0x400B 8030
MCMAT1	Match register, channel 1	R/W	0xFFFF FFFF	0x400B 8034
MCMAT2	Match register, channel 2	R/W	0xFFFF FFFF	0x400B 8038
MCDT	Dead time register	R/W	0x3FFF FFFF	0x400B 803C
MCCP	Commutation Pattern register	R/W	0	0x400B 8040
MCCAP0	Capture register, channel 0	RO	0	0x400B 8044
MCCAP1	Capture register, channel 1	RO	0	0x400B 8048
MCCAP2	Capture register, channel 2	RO	0	0x400B 804C
MCINTEN	Interrupt Enable read address	RO	0	0x400B 8050
MCINTEN_SET	Interrupt Enable set address	WO	-	0x400B 8054
MCINTEN_CLR	Interrupt Enable clear address	WO	-	0x400B 8058
MCCNTCON	Count Control read address	RO	0	0x400B 805C
MCCNTCON_SET	Count Control set address	WO	-	0x400B 8060
MCCNTCON_CLR	Count Control clear address	WO	-	0x400B 8064
MCINTF	Interrupt flags read address	RO	0	0x400B 8068
MCINTF_SET	Interrupt flags set address	WO	-	0x400B 806C
MCINTF_CLR	Interrupt flags clear address	WO	-	0x400B 8070
MCCAP_CLR	Capture clear address	WO	-	0x400B 8074

25.7.1 MCPWM Control register

25.7.1.1 MCPWM Control read address (MCCON - 0x400B 8000)

The MCCON register controls the operation of all channels of the PWM. This address is read-only, but the underlying register can be modified by writing to addresses MCCON_SET and MCCON_CLR.

Table 456. MCPWM Control read address (MCCON - 0x400B 8000) bit description

Bit	Symbol	Value	Description	Reset value
0	RUN0		Stops/starts timer channel 0.	0
		0	Stop.	
		1	Run.	
1	CENTER0		Edge/center aligned operation for channel 0.	0
		0	Edge-aligned.	
		1	Center-aligned.	
2	POLA0		Selects polarity of the MCOA0 and MCOB0 pins.	0
		0	Passive state is LOW, active state is HIGH.	
		1	Passive state is HIGH, active state is LOW.	
3	DTE0		Controls the dead-time feature for channel 0.	0
		0	Dead-time disabled.	
		1	Dead-time enabled.	
4	DISUP0		Enable/disable updates of functional registers for channel 0 (see Section 25.8.2, Section 25.7.6.1).	0
		0	Functional registers are updated from the write registers at the end of each PWM cycle.	
		1	Functional registers remain the same as long as the timer is running.	
7:5	-	-	Reserved.	
8	RUN1		Stops/starts timer channel 1.	0
		0	Stop.	
		1	Run.	
9	CENTER1		Edge/center aligned operation for channel 1.	0
		0	Edge-aligned.	
		1	Center-aligned.	
10	POLA1		Selects polarity of the MCOA1 and MCOB1 pins.	0
		0	Passive state is LOW, active state is HIGH.	
		1	Passive state is HIGH, active state is LOW.	
11	DTE1		Controls the dead-time feature for channel 1.	0
		0	Dead-time disabled.	
		1	Dead-time enabled.	
12	DISUP1		Enable/disable updates of functional registers for channel 1 (see Section 25.8.2, Section 25.7.6.1).	0
		0	Functional registers are updated from the write registers at the end of each PWM cycle.	
		1	Functional registers remain the same as long as the timer is running.	

Table 456. MCPWM Control read address (MCCON - 0x400B 8000) bit description *…continued*

Bit	Symbol	Value	Description	Reset value
15:13	-	-	Reserved.	0
16	RUN2		Stops/starts timer channel 2.	0
		0	Stop.	
		1	Run.	
17	CENTER2		Edge/center aligned operation for channel 2.	0
		0	Edge-aligned.	
		1	Center-aligned.	
18	POLA2		Selects polarity of the MCOA2 and MCOB2 pins.	0
		0	Passive state is LOW, active state is HIGH.	
		1	Passive state is HIGH, active state is LOW.	
19	DTE2		Controls the dead-time feature for channel 1.	0
		0	Dead-time disabled.	
		1	Dead-time enabled.	
20	DISUP2		Enable/disable updates of functional registers for channel 2 (see Section 25.8.2, Section 25.7.6.1).	0
		0	Functional registers are updated from the write registers at the end of each PWM cycle.	
		1	Functional registers remain the same as long as the timer is running.	
28:21	-	-	Reserved.	
29	INVBDC		Controls the polarity of the MCOB outputs for all 3 channels. This bit is typically set to 1 only in 3-phase DC mode.	
		0	The MCOB outputs have opposite polarity from the MCOA outputs (aside from dead time).	
		1	The MCOB outputs have the same basic polarity as the MCOA outputs. (see Section 25.8.6)	
30	ACMODE		3-phase AC mode select (see Section 25.8.7).	0
		0	3-phase AC-mode off: Each PWM channel uses its own timer-counter and period register.	
		1	3-phase AC-mode on: All PWM channels use the timer-counter and period register of channel 0.	
31	DCMODE		3-phase DC mode select (see Section 25.8.6).	0
		0	3-phase DC mode off: PWM channels are independent (unless bit ACMODE = 1)	
		1	3-phase DC mode on: The internal MCOA0 output is routed through the MCCP (i.e. a mask) register to all six PWM outputs.	

25.7.1.2 MCPWM Control set address (MCCON_SET - 0x400B 8004)

Writing ones to this write-only address sets the corresponding bits in MCCON.

Table 457. MCPWM Control set address (MCCON_SET - 0x400B 8004) bit description

Bit	Description
31:0	Writing ones to this address sets the corresponding bits in the MCCON register. See Table 456.

25.7.1.3 MCPWM Control clear address (MCCON_CLR - 0x400B 8008)

Writing ones to this write-only address clears the corresponding bits in MCCON.

Table 458. MCPWM Control clear address (MCCON_CLR - 0x400B 8008) bit description

Bit	Description
31:0	Writing ones to this address clears the corresponding bits in the MCCON register. See Table 456.

25.7.2 MCPWM Capture Control register

25.7.2.1 MCPWM Capture Control read address (MCCAPCON - 0x400B 800C)

The MCCAPCON register controls detection of events on the MCI0-2 inputs for all MCPWM channels. Any of the three MCI inputs can be used to trigger a capture event on any or all of the three channels. This address is read-only, but the underlying register can be modified by writing to addresses MCCAPCON_SET and MCCAPCON_CLR.

Table 459. MCPWM Capture Control read address (MCCAPCON - 0x400B 800C) bit description

Bit	Symbol	Description	Reset Value
0	CAP0MCI0_RE	A 1 in this bit enables a channel 0 capture event on a rising edge on MCI0.	0
1	CAP0MCI0_FE	A 1 in this bit enables a channel 0 capture event on a falling edge on MCI0.	0
2	CAP0MCI1_RE	A 1 in this bit enables a channel 0 capture event on a rising edge on MCI1.	0
3	CAP0MCI1_FE	A 1 in this bit enables a channel 0 capture event on a falling edge on MCI1.	0
4	CAP0MCI2_RE	A 1 in this bit enables a channel 0 capture event on a rising edge on MCI2.	0
5	CAP0MCI2_FE	A 1 in this bit enables a channel 0 capture event on a falling edge on MCI2.	0
6	CAP1MCI0_RE	A 1 in this bit enables a channel 1 capture event on a rising edge on MCI0.	0
7	CAP1MCI0_FE	A 1 in this bit enables a channel 1 capture event on a falling edge on MCI0.	0
8	CAP1MCI1_RE	A 1 in this bit enables a channel 1 capture event on a rising edge on MCI1.	0
9	CAP1MCI1_FE	A 1 in this bit enables a channel 1 capture event on a falling edge on MCI1.	0
10	CAP1MCI2_RE	A 1 in this bit enables a channel 1 capture event on a rising edge on MCI2.	0
11	CAP1MCI2_FE	A 1 in this bit enables a channel 1 capture event on a falling edge on MCI2.	0
12	CAP2MCI0_RE	A 1 in this bit enables a channel 2 capture event on a rising edge on MCI0.	0
13	CAP2MCI0_FE	A 1 in this bit enables a channel 2 capture event on a falling edge on MCI0.	0
14	CAP2MCI1_RE	A 1 in this bit enables a channel 2 capture event on a rising edge on MCI1.	0
15	CAP2MCI1_FE	A 1 in this bit enables a channel 2 capture event on a falling edge on MCI1.	0
16	CAP2MCI2_RE	A 1 in this bit enables a channel 2 capture event on a rising edge on MCI2.	0
17	CAP2MCI2_FE	A 1 in this bit enables a channel 2 capture event on a falling edge on MCI2.	0
18	RT0	If this bit is 1, TC0 is reset by a channel 0 capture event.	0
19	RT1	If this bit is 1, TC1 is reset by a channel 1 capture event.	0
20	RT2	If this bit is 1, TC2 is reset by a channel 2 capture event.	0
21	HNFCAP0	Hardware noise filter: if this bit is 1, channel 0 capture events are delayed as described in Section 25.8.4.	0
22	HNFCAP1	Hardware noise filter: if this bit is 1, channel 1 capture events are delayed as described in Section 25.8.4.	0
23	HNFCAP2	Hardware noise filter: if this bit is 1, channel 2 capture events are delayed as described in Section 25.8.4.	0
31:24	-	Reserved.	-

25.7.2.2 MCPWM Capture Control set address (MCCAPCON_SET - 0x400B 8010)

Writing ones to this write-only address sets the corresponding bits in MCCAPCON.

Table 460. MCPWM Capture Control set address (MCCAPCON_SET - 0x400B 8010) bit description

Bit	Description
31:0	Writing ones to this address sets the corresponding bits in the MCCAPCON register. See Table 459.

25.7.2.3 MCPWM Capture control clear address (MCCAPCON_CLR - 0x400B 8014)

Writing ones to this write-only address clears the corresponding bits in MCCAPCON.

Table 461. MCPWM Capture control clear register (MCCAPCON_CLR - address 0x400B 8014) bit description

Bit	Description
31:0	Writing ones to this address clears the corresponding bits in the MCCAPCON register. See Table 459.

25.7.3 MCPWM Interrupt registers

The Motor Control PWM module includes the following interrupt sources:

Table 462. Motor Control PWM interrupts

Symbol	Description
ILIM0/1/2	Limit interrupts for channels 0, 1, 2.
IMAT0/1/2	Match interrupts for channels 0, 1, 2.
ICAP0/1/2	Capture interrupts for channels 0, 1, 2.
ABORT	Fast abort interrupt

All MCPWM interrupt registers contain one bit for each source as shown in Table 463.

Table 463. Interrupt sources bit allocation table

Bit	31	30	29	28	27	26	25	24
Symbol	-	-	-	-	-	-	-	-
Bit	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**
Symbol	-	-	-	-	-	-	-	-
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Symbol	ABORT	-	-	-	-	ICAP2	IMAT2	ILIM2
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Symbol	-	ICAP1	IMAT1	ILIM1	-	ICAP0	IMAT0	ILIM0

7.3.1 MCPWM Interrupt Enable read address (MCINTEN - 0x400B 8050)

The MCINTEN register controls which of the MCPWM interrupts are enabled. This address is read-only, but the underlying register can be modified by writing to addresses MCINTEN_SET and MCINTEN_CLR.

Table 464. MCPWM Interrupt Enable read address (MCINTEN - 0x400B 8050) bit description

Bit	Value	Description	Reset value
31:0		See Table 463 for the bit allocation.	0
	1	Interrupt enabled.	
	0	Interrupt disabled.	

25.7.3.2 MCPWM Interrupt Enable set address (MCINTEN_SET - 0x400B 8054)

Writing ones to this write-only address sets the corresponding bits in MCINTEN, thus enabling interrupts.

Table 465. PWM interrupt enable set register (MCINTEN_SET - address 0x400B 8054) bit description

Bit	Description
31:0	Writing ones to this address sets the corresponding bits in MCINTEN, thus enabling interrupts. See Table 463.

25.7.3.3 MCPWM Interrupt Enable clear address (MCINTEN_CLR - 0x400B 8058)

Writing ones to this write-only address clears the corresponding bits in MCINTEN, thus disabling interrupts.

Table 466. PWM interrupt enable clear register (MCINTEN_CLR - address 0x400B 8058) bit description

Bit	Description
31:0	Writing ones to this address clears the corresponding bits in MCINTEN, thus disabling interrupts. See Table 463.

25.7.3.4 MCPWM Interrupt Flags read address (MCINTF - 0x400B 8068)

The MCINTF register includes all MCPWM interrupt flags, which are set when the corresponding hardware event occurs, or when ones are written to the MCINTF_SET address. When corresponding bits in this register and MCINTEN are both 1, the MCPWM asserts its interrupt request to the Interrupt Controller module. This address is read-only, but the bits in the underlying register can be modified by writing ones to addresses MCINTF_SET and MCINTF_CLR.

Table 467. MCPWM Interrupt Flags read address (MCINTF - 0x400B 8068) bit description

Bit	Value	Description	Reset value
31:0		See Table 463 for the bit allocation.	0
	1	If the corresponding bit in MCINTEN is 1, the MCPWM module is asserting its interrupt request to the Interrupt Controller.	
	0	This interrupt source is not contributing to the MCPWM interrupt request.	

25.7.3.5 MCPWM Interrupt Flags set address (MCINTF_SET - 0x400B 806C)

Writing one(s) to this write-only address sets the corresponding bit(s) in MCINTF, thus possibly simulating hardware interrupt(s).

Table 468. MCPWM Interrupt Flags set address (PWMINTF_SET - 0x400B 806C) bit description

Bit	Description
31:0	Writing one(s) to this write-only address sets the corresponding bit(s) in the MCINTF register, thus possibly simulating hardware interrupt(s). See Table 463.

25.7.3.6 MCPWM Interrupt Flags clear address (MCINTF_CLR - 0x400B 8070)

Writing one(s) to this write-only address sets the corresponding bit(s) in MCINTF, thus clearing the corresponding interrupt request(s). This is typically done in interrupt service routines.

Table 469. MCPWM Interrupt Flags clear address (PWMINTF_CLR - 0x400B 8070) bit description

Bit	Description
31:0	Writing one(s) to this write-only address sets the corresponding bit(s) in the MCINTF register, thus clearing the corresponding interrupt request(s). See Table 463.

25.7.4 MCPWM Count Control register

25.7.4.1 MCPWM Count Control read address (MCCNTCON - 0x400B 805C)

The MCCNTCON register controls whether the MCPWM channels are in timer or counter mode, and in counter mode whether the counter advances on rising and/or falling edges on any or all of the three MCI inputs. If timer mode is selected, the counter advances based on the PCLK clock.

This address is read-only. To set or clear the register bits, write ones to the MCCNTCON_SET or MCCNTCON_CLR address.

Table 470. MCPWM Count Control read address (MCCNTCON - 0x400B 805C) bit description

Bit	Symbol	Value	Description	Reset Value
0	TC0MCI0_RE	1	If MODE0 is 1, counter 0 advances on a rising edge on MCI0.	0
		0	A rising edge on MCI0 does not affect counter 0.	
1	TC0MCI0_FE	1	If MODE0 is 1, counter 0 advances on a falling edge on MCI0.	0
		0	A falling edge on MCI0 does not affect counter 0.	
2	TC0MCI1_RE	1	If MODE0 is 1, counter 0 advances on a rising edge on MCI1.	0
		0	A rising edge on MCI1 does not affect counter 0.	
3	TC0MCI1_FE	1	If MODE0 is 1, counter 0 advances on a falling edge on MCI1.	0
		0	A falling edge on MCI1 does not affect counter 0.	
4	TC0MCI2_RE	1	If MODE0 is 1, counter 0 advances on a rising edge on MCI2.	0
		0	A rising edge on MCI0 does not affect counter 0.	
5	TC0MCI2_FE	1	If MODE0 is 1, counter 0 advances on a falling edge on MCI2.	0
		0	A falling edge on MCI0 does not affect counter 0.	
6	TC1MCI0_RE	1	If MODE1 is 1, counter 1 advances on a rising edge on MCI0.	0
		0	A rising edge on MCI0 does not affect counter 1.	
7	TC1MCI0_FE	1	If MODE1 is 1, counter 1 advances on a falling edge on MCI0.	0
		0	A falling edge on MCI0 does not affect counter 1.	
8	TC1MCI1_RE	1	If MODE1 is 1, counter 1 advances on a rising edge on MCI1.	0
		0	A rising edge on MCI1 does not affect counter 1.	
9	TC1MCI1_FE	1	If MODE1 is 1, counter 1 advances on a falling edge on MCI1.	0
		0	A falling edge on MCI0 does not affect counter 1.	
10	TC1MCI2_RE	1	If MODE1 is 1, counter 1 advances on a rising edge on MCI2.	0
		0	A rising edge on MCI2 does not affect counter 1.	
11	TC1MCI2_FE	1	If MODE1 is 1, counter 1 advances on a falling edge on MCI2.	0
		0	A falling edge on MCI2 does not affect counter 1.	
12	TC2MCI0_RE	1	If MODE2 is 1, counter 2 advances on a rising edge on MCI0.	0
		0	A rising edge on MCI0 does not affect counter 2.	

Table 470. MCPWM Count Control read address (MCCNTCON - 0x400B 805C) bit description *…continued*

Bit	Symbol	Value	Description	Reset Value
13	TC2MCI0_FE	1	If MODE2 is 1, counter 2 advances on a falling edge on MCI0.	0
		0	A falling edge on MCI0 does not affect counter 2.	
14	TC2MCI1_RE	1	If MODE2 is 1, counter 2 advances on a rising edge on MCI1.	0
		0	A rising edge on MCI1 does not affect counter 2.	
15	TC2MCI1_FE	1	If MODE2 is 1, counter 2 advances on a falling edge on MCI1.	0
		0	A falling edge on MCI1 does not affect counter 2.	
16	TC2MCI2_RE	1	If MODE2 is 1, counter 2 advances on a rising edge on MCI2.	0
		0	A rising edge on MCI2 does not affect counter 2.	
17	TC2MCI2_FE	1	If MODE2 is 1, counter 2 advances on a falling edge on MCI2.	0
		0	A falling edge on MCI2 does not affect counter 2.	
28:18	-	-	Reserved.	-
29	CNTR0	1	Channel 0 is in counter mode.	0
		0	Channel 0 is in timer mode.	
30	CNTR1	1	Channel 1 is in counter mode.	0
		0	Channel 1 is in timer mode.	
31	CNTR2	1	Channel 2 is in counter mode.	0
		0	Channel 2 is in timer mode.	

25.7.4.2 MCPWM Count Control set address (MCCNTCON_SET - 0x400B 8060)

Writing one(s) to this write-only address sets the corresponding bit(s) in MCCNTCON.

Table 471. MCPWM Count Control set address (MCCNTCON_SET - 0x400B 8060) bit description

Bit	Description
31:0	Writing one(s) to this write-only address sets the corresponding bit(s) in the MCCNTCON register. See Table 470.

25.7.4.3 MCPWM Count Control clear address (MCCNTCON_CLR - 0x400B 8064)

Writing one(s) to this write-only address clears the corresponding bit(s) in MCCNTCON.

Table 472. MCPWM Count Control clear address (MCCAPCON_CLR - 0x400B 8064) bit description

Bit	Description
31:0	Writing one(s) to this write-only address clears the corresponding bit(s) in the MCCNTCON register. See Table 470.

25.7.5 MCPWM Timer/Counter 0-2 registers (MCTC0-2 - 0x400B 8018, 0x400B 801C, 0x400B 8020)

These registers hold the current values of the 32-bit counter/timers for channels 0-2. Each value is incremented on every PCLK, or by edges on the MCI0-2 pins, as selected by MCCNTCON. The timer/counter counts up from 0 until it reaches the value in its corresponding MCPER register (or is stopped by writing to MCCON_CLR).

A TC register can be read at any time. In order to write to the TC register, its channel must be stopped. If not, the write will not take place, no exception is generated.

Table 473. MCPWM Timer/Counter 0-2 registers (MCTC0-2 - 0x400B 8018, 0x400B 801C, 0x400B 8020) bit description

Bit	Symbol	Description	Reset value
31:0	MCTC0/1/2	Timer/Counter values for channels 0, 1, 2.	0

25.7.6 MCPWM Limit 0-2 registers (MCLIM0-2 - 0x400B 8024, 0x400B 8028, 0x400B 802C)

These registers hold the limiting values for timer/counters 0-2. When a timer/counter reaches its corresponding limiting value: 1) in edge-aligned mode, it is reset and starts over at 0; 2) in center-aligned mode, it begins counting down until it reaches 0, at which time it begins counting up again.

If the channel's CENTER bit in MCCON is 0 selecting edge-aligned mode, the match between TC and LIM switches the channel's A output from "active" to "passive" state. If the channel's CENTER and DTE bits in MCCON are both 0, the match simultaneously switches the channel's B output from "passive" to "active" state.

If the channel's CENTER bit is 0 but the DTE bit is 1, the match triggers the channel's deadtime counter to begin counting -- when the deadtime counter expires, the channel's B output switches from "passive" to "active" state.

In center-aligned mode, matches between a channel's TC and LIM registers have no effect on its A and B outputs.

Table 474. MCPWM Limit 0-2 registers (MCLIM0-2 - 0x400B 8024, 0x400B 8028, 0x400B 802C) bit description

Bit	Symbol	Description	Reset value
31:0	MCLIM0/1/2	Limit values for TC0, 1, 2.	0xFFFF FFFF

25.7.6.1 Match and Limit write and operating registers

Writing to either a Limit or a Match register loads a "write" register, and, if the channel is stopped, it also loads a shadow "operating" register that is compared to the TC. If the channel is running and its "disable update" bit in MCCON is 0, the operating registers are loaded from the write registers as follows:

1. in edge-aligned mode, when the TC matches the operating Limit register;
2. in center-aligned mode, when the TC counts back down to 0. If the channel is running and the "disable update" bit is 1, the operating registers are not loaded from the write registers until software stops the channel.

Reading an MCLIM register address always returns the operating value.

Simultaneous update of the Limit and Match registers can be achieved by writing the MCLIMx and MCMATx registers and then clearing the DISUPx bits in the MCCON register (see Table 458). The simultaneous update will occur at the beginning of the next PWM cycle (also see Section 25.8.2).

Remark: In timer mode, the period of a channel's modulated MCO outputs is determined by its Limit register, and the pulse width at the start of the period is determined by its Match register. You can consider the Limit register to be a "Period register" and the Match register to be a "Pulse Width register".

25.7.7 MCPWM Match 0-2 registers (MCMAT0-2 - 0x400B 8030, 0x400B 8034, 0x400B 8038)

These registers also have “write” and “operating” versions as described above for the Limit registers, and the operating registers are also compared to the channels’ TCs. See 25.7.6 above for details of reading and writing both Limit and Match registers.

The Match and Limit registers control the MCO0-2 outputs. If a Match register is to have any effect on its channel’s operation, it must contain a smaller value than the corresponding Limit register.

Table 475. MCPWM Match 0-2 registers (MCMAT0-2 - addresses 0x400B 8030, 0x400B 8034, 0x400B 8038) bit description

Bit	Symbol	Description	Reset value
31:0	MCMAT0/1/2	Match values for TC0, 1, 2.	0xFFFF FFFF

25.7.7.1 Match register in Edge-Aligned mode

If the channel’s CENTER bit in MCCON is 0 selecting edge-aligned mode, a match between TC and MAT switches the channel’s B output from “active” to “passive” state. If the channel’s CENTER and DTE bits in MCCON are both 0, the match simultaneously switches the channel’s A output from “passive” to “active” state.

If the channel’s CENTER bit is 0 but the DTE bit is 1, the match triggers the channel’s deadtime counter to begin counting -- when the deadtime counter expires, the channel’s A output switches from “passive” to “active” state.

25.7.7.2 Match register in Center-Aligned mode

If the channel’s CENTER bit in MCCON is 1 selecting center-aligned mode, a match between TC and MAT while the TC is incrementing switches the channel’s B output from “active” to “passive” state, and a match while the TC is decrementing switches the A output from “active” to “passive”. If the channel’s CENTER bit in MCCON is 1 but the DTE bit is 0, a match simultaneously switches the channel’s other output in the opposite direction.

If the channel’s CENTER and DTE bits are both 1, a match between TC and MAT triggers the channel’s deadtime counter to begin counting -- when the deadtime counter expires, the channel’s B output switches from “passive” to “active” if the TC was counting up at the time of the match, and the channel’s A output switches from “passive” to “active” if the TC was counting down at the time of the match.

25.7.7.3 0 and 100% duty cycle

To lock a channel’s MCO outputs at the state “B active, A passive”, write its Match register with a higher value than you write to its Limit register. The match never occurs.

To lock a channel’s MCO outputs at the opposite state, “A active, B passive”, simply write 0 to its Match register.

25.7.8 MCPWM Dead-time register (MCDT - 0x400B 803C)

This register holds the dead-time values for the three channels. If a channel's DTE bit in MCCON is 1 to enable its dead-time counter, the counter counts down from this value whenever one its channel's outputs changes from "active" to "passive" state. When the dead-time counter reaches 0, the channel changes its other output from "passive" to "active" state.

The motivation for the dead-time feature is that power transistors, like those driven by the A and B outputs in a motor-control application, take longer to fully turn off than they take to start to turn on. If the A and B transistors are ever turned on at the same time, a wasteful and damaging current will flow between the power rails through the transistors. In such applications, the dead-time register should be programmed with the number of PCLK periods that is greater than or equal to the transistors' maximum turn-off time minus their minimum turn-on time.

Table 476. MCPWM Dead-time register (MCDT - address 0x400B 803C) bit description

Bit	Symbol	Description	Reset value
9:0	DT0	Dead time for channel 0.[1]	0x3FF
19:10	DT1	Dead time for channel 1.[2]	0x3FF
29:20	DT2	Dead time for channel 2.[2]	0x3FF
31:30	-	reserved	

[1] If ACMODE is 1 selecting AC-mode, this field controls the dead time for all three channels.

[2] If ACMODE is 0.

25.7.9 MCPWM Commutation Pattern register (MCCP - 0x400B 8040)

This register is used in DC mode only. The internal MCOA0 signal is routed to any or all of the six output pins under the control of the bits in this register. Like the Match and Limit registers, this register has "write" and "operational" versions. See 25.7.6 and 25.8.2 for more about this subject.

Table 477. MCPWM Commutation Pattern register (MCCP - address 0x400B 8040) bit description

Bit	Symbol	Description	Reset value
0	CCPA0	0 = MCOA0 passive, 1 = internal MCOA0.	0
1	CCPB0	0 = MCOB0 passive, 1 = MCOB0 tracks internal MCOA0.	0
2	CCPA1	0 = MCOA1 passive, 1 = MCOA1 tracks internal MCOA0.	0
3	CCPB1	0 = MCOB1 passive, 1 = MCOB1 tracks internal MCOA0.	0
4	CCPA2	0 = MCOA2 passive, 1 = MCOA2 tracks internal MCOA0.	0
5	CCPB2	0 = MCOB2 passive, 1 = MCOB2 tracks internal MCOA0.	0
31:6	-	Reserved.	

25.7.10 MCPWM Capture Registers

25.7.10.1 MCPWM Capture read addresses (MCCAP0-2 - 0x400B 8044, 0x400B 8048, 0x400B 804C)

The MCCAPCON register (Table 459) allows software to select any edge(s) on any of the MCI0-2 inputs as a capture event for each channel. When a channel's capture event occurs, the current TC value for that channel is stored in its read-only Capture register. These addresses are read-only, but the underlying registers can be cleared by writing to the CAP_CLR address

Table 478. MCPWM Capture read addresses (MCCAP0/1/2 - 0x400B 8044, 0x400B 8048, 0x400B 804C) bit description

Bit	Symbol	Description	Reset value
31:0	CAP0/1/2	TC value at a capture event for channels 0, 1, 2.	0x0000 0000

25.7.10.2 MCPWM Capture clear address (MCCAP_CLR - 0x400B 8074)

Writing ones to this write-only address clears the selected CAP register(s).

Table 479. MCPWM Capture clear address (CAP_CLR - 0x400B 8074) bit description

Bit	Symbol	Description
0	CAP_CLR0	Writing a 1 to this bit clears the MCCAP0 register.
1	CAP_CLR1	Writing a 1 to this bit clears the MCCAP1 register.
2	CAP_CLR2	Writing a 1 to this bit clears the MCCAP2 register.
31:3	-	Reserved

25.8 PWM operation

25.8.1 Pulse-width modulation

Each channel of the MCPWM has two outputs, A and B, that can drive a pair of transistors to switch a controlled point between two power rails. Most of the time the two outputs have opposite polarity, but a dead-time feature can be enabled (on a per-channel basis) to delay both signals' transitions from “passive” to “active” state so that the transistors are never both turned on simultaneously. In a more general view, the states of each output pair can be thought of “high”, “low”, and “floating” or “up”, “down”, and “center-off”.

Each channel's mapping from “active” and “passive” to “high” and “low” is programmable. After Reset, the three A outputs are passive/low, and the B outputs are active/high.

The MCPWM can perform edge-aligned and center-aligned pulse-width modulation.

Note: *In timer mode, the period of a channel's modulated MCO outputs is determined by its Limit register, and the pulse width at the start of the period is determined by its Match register. If it suits your way of thinking, consider the Limit register to be the “Period register” and the Match register to be the “Pulse Width register”.*

Edge-aligned PWM without dead-time

In this mode the timer TC counts up from 0 to the value in the LIM register. As shown in Figure 123, the MCO state is “A passive” until the TC matches the Match register, at which point it changes to “A active”. When the TC matches the Limit register, the MCO state changes back to “A passive”, and the TC is reset and starts counting up again.

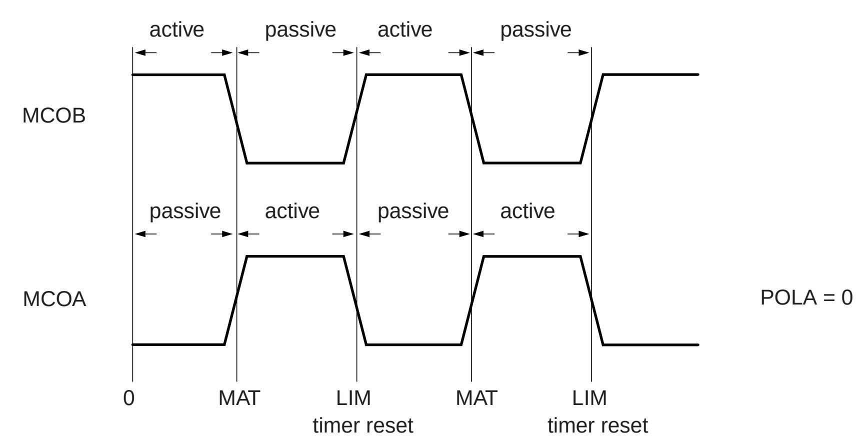

Fig 123. Edge-aligned PWM waveform without dead time, POLA = 0

Center-aligned PWM without dead-time

In this mode the timer TC counts up from 0 to the value in the LIM register, then counts back down to 0 and repeats. As shown in Figure 124, while the timer counts up, the MCO state is “A passive” until the TC matches the Match register, at which point it changes to “A active”. When the TC matches the Limit register it starts counting down. When the TC matches the Match register on the way down, the MCO state changes back to “A passive”.

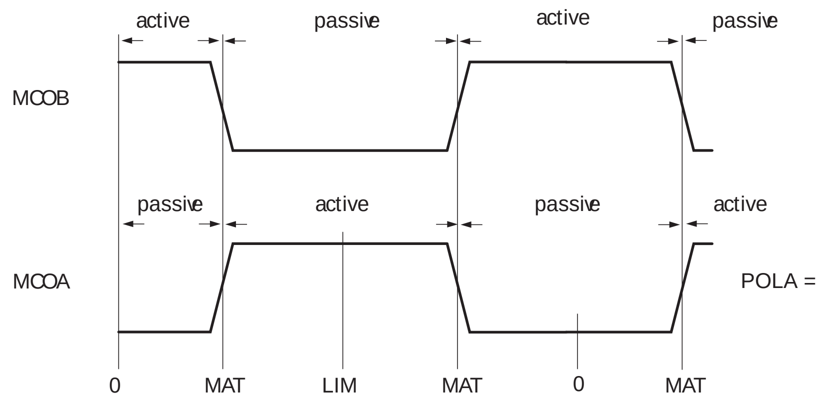

Fig 124. Center-aligned PWM waveform without dead time, POLA = 0

Dead-time counter

When the a channel's DTE bit is set in MCCON, the dead-time counter delays the passive-to-active transitions of both MCO outputs. The dead-time counter starts counting down, from the channel's DT value (in the MCDT register) to 0, whenever the channel's A or B output changes from active to passive. The transition of the other output from passive to active is delayed until the dead-time counter reaches 0. During the dead time, the MCOA and MCOB output levels are both passive. Figure 125 shows operation in edge aligned mode with dead time, and Figure 126 shows center-aligned operation with dead time.

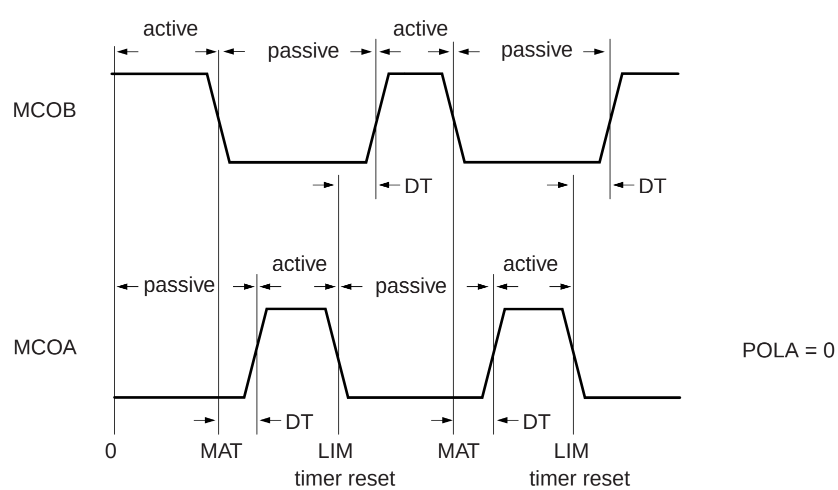

Fig 125. Edge-aligned PWM waveform with dead time, POLA = 0

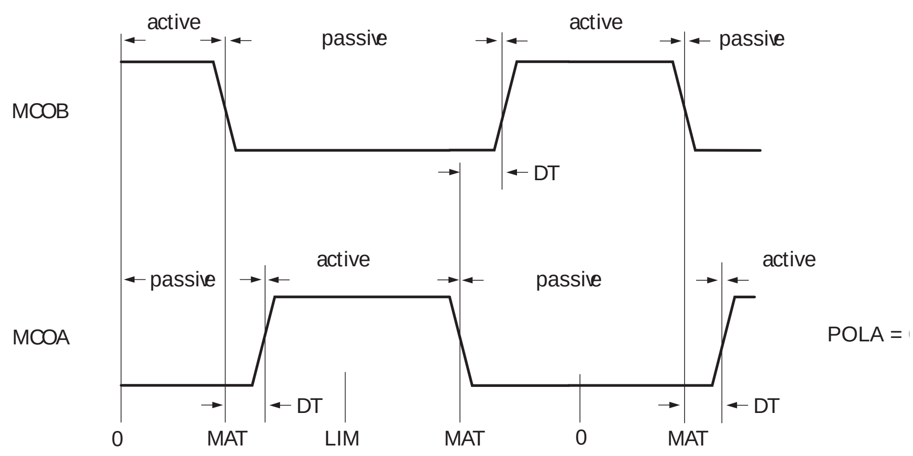

Fig 126. Center-aligned waveform with dead time, POLA = 0

25.8.2 Shadow registers and simultaneous updates

The Limit, Match, and Commutation Pattern registers (MCLIM, MCMAT, and MCCP) are implemented as register pairs, each consisting of a write register and an operational register. Software writes into the write registers. The operational registers control the actual operation of each channel and are loaded with the current value in the write registers when the TC starts counting up from 0.

Updating of the functional registers can be disabled by setting a channel's DISUP bit in the MCCON register. If the DISUP bits are set, the functional registers are not updated until software stops the channel.

If a channel is not running when software writes to its LIM or MAT register, the functional register is updated immediately.

Software can write to a TC register only when its channel is stopped.

25.8.3 Fast Abort (ABORT)

The MCPWM has an external input $\overline{\text{MCABORT}}$. When this input goes low, all six MCO outputs assume their "A passive" states, and the Abort interrupt is generated if enabled. The outputs remain locked in "A passive" state until the ABORT interrupt flag is cleared or the Abort interrupt is disabled. The ABORT flag may not be cleared before the $\overline{\text{MCABORT}}$ input goes high.

In order to clear an ABORT flag, a 1 must be written to bit 15 of the MCINTF_CLR register. This will remove the interrupt request. The interrupt can also be disabled by writing a 1 to bit 15 of the MCINTEN_CLR register.

25.8.4 Capture events

Each PWM channel can take a snapshot of its TC when an input signal transitions. Any channel may use any combination of rising and/or falling edges on any or all of the MCI0-2 inputs as a capture event, under control of the MCCAPCON register. Rising or falling edges on the inputs are detected synchronously with respect to PCLK.

If a channel's HNF bit in the MCCAPCON register is set to enable “noise filtering”, a selected edge on an MCI pin starts the dead-time counter for that channel, and the capture event actions described below are delayed until the dead-time counter reaches 0. This function is targeted specifically for performing three-phase brushless DC motor control with Hall sensors.

A capture event on a channel (possibly delayed by HNF) causes the following:

- The current value of the TC is stored in the Capture register (CAP).
- If the channel's capture event interrupt is enabled (see Table 464), the capture event interrupt flag is set.
- If the channel's RT bit is set in the MCCAPCON register, enabling reset on a capture event, the input event has the same effect as matching the channel's TC to its LIM register. This includes resetting the TC and switching the MCO pin(s) in edge-aligned mode as described in 25.7.6 and 25.8.1.

25.8.5 External event counting (Counter mode)

If a channel's MODE bit is 1 in MCCNTCON, its TC is incremented by rising and/or falling edge(s) (synchronously detected) on the MCI0-2 input(s), rather than by PCLK. The PWM functions and capture functions are unaffected.

25.8.6 Three-phase DC mode

The three-phase DC mode is selected by setting the DCMODE bit in the MCCON register.

In this mode, the internal MCOA0 signal can be routed to any or all of the MCO outputs. Each MCO output is masked by a bit in the current Commutation Pattern register MCCP. If a bit in the MCCP register is 0, its output pin has the logic level for the passive state of output MCOA0. The polarity of the off state is determined by the POLA0 bit.

All MCO outputs that have 1 bits in the MCCP register are controlled by the internal MCOA0 signal.

The three MCOB output pins are inverted when the INVBDC bit is 1 in the MCCON register. This feature accommodates bridge-drivers that have active-low inputs for the low-side switches.

The MCCP register is implemented as a shadow register pair, so that changes to the active commutation pattern occur at the beginning of a new PWM cycle. See 25.7.6 and 25.8.2 for more about writing and reading such registers.

Figure 127 shows sample waveforms of the MCO outputs in three-phase DC mode. Bits 1 and 3 in the MCCP register (corresponding to outputs MCOB1 and MCOB0) are set to 0 so that these outputs are masked and in the off state. Their logic level is determined by the POLA0 bit (here, POLA0 = 0 so the passive state is logic LOW). The INVBDC bit is set to 0 (logic level not inverted) so that the B output have the same polarity as the A outputs. Note that this mode differs from other modes in that the MCOB outputs are **not** the opposite of the MCOA outputs.

In the situation shown in Figure 127, bits 0, 2, 4, and 5 in the MCCP register are set to 1. That means that MCOA1 and both MCO outputs for channel 2 follow the MCOA0 signal.

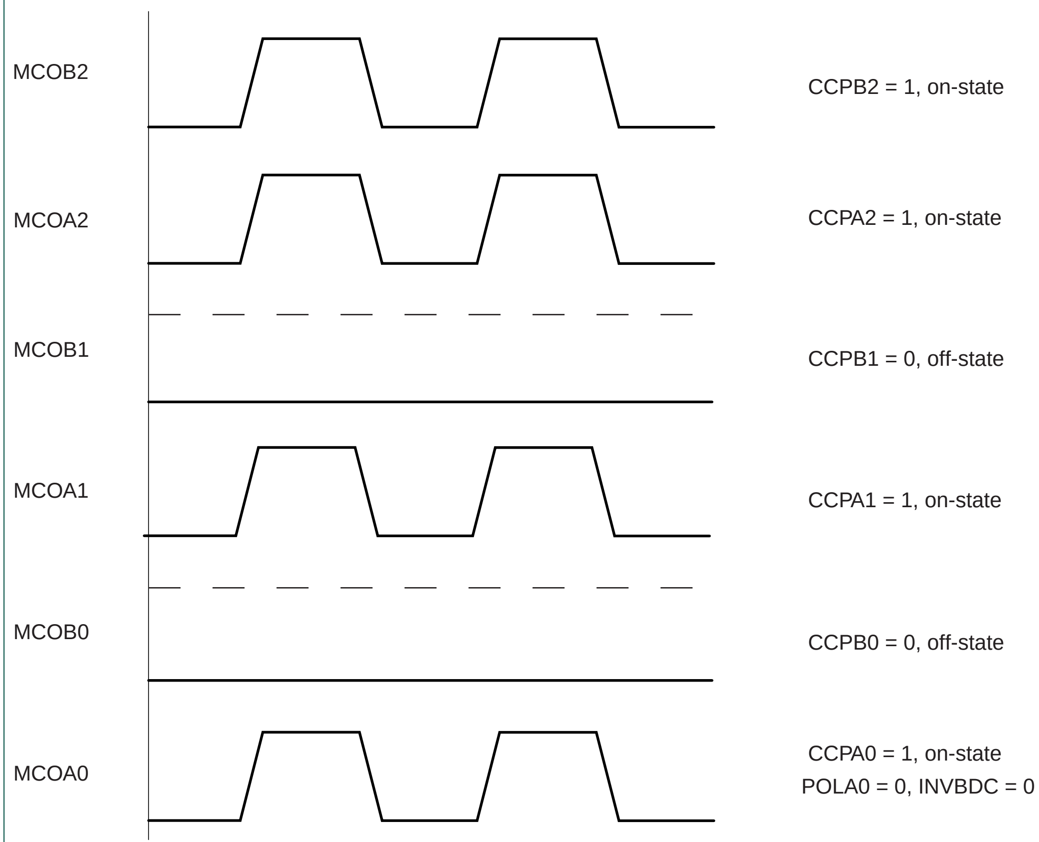

Fig 127. Three-phase DC mode sample waveforms

25.8.7 Three phase AC mode

The three-phase AC-mode is selected by setting the ACMODE bit in the MCCON register.

In this mode, the value of channel 0's TC is routed to all channels for comparison with their MAT registers. (The LIM1-2 registers are not used.)

Each channel controls its MCO output by comparing its MAT value to TC0.

Figure 128 shows sample waveforms for the six MCO outputs in three-phase AC mode. The POLA bits are set to 0 for all three channels, so that for all MCO outputs the active levels are high and the passive levels are low. Each channel has a different MAT value which is compared to the MCTC0 value. In this mode the period value is identical for all three channels and is determined by MCLIM0. The dead-time mode is disabled.

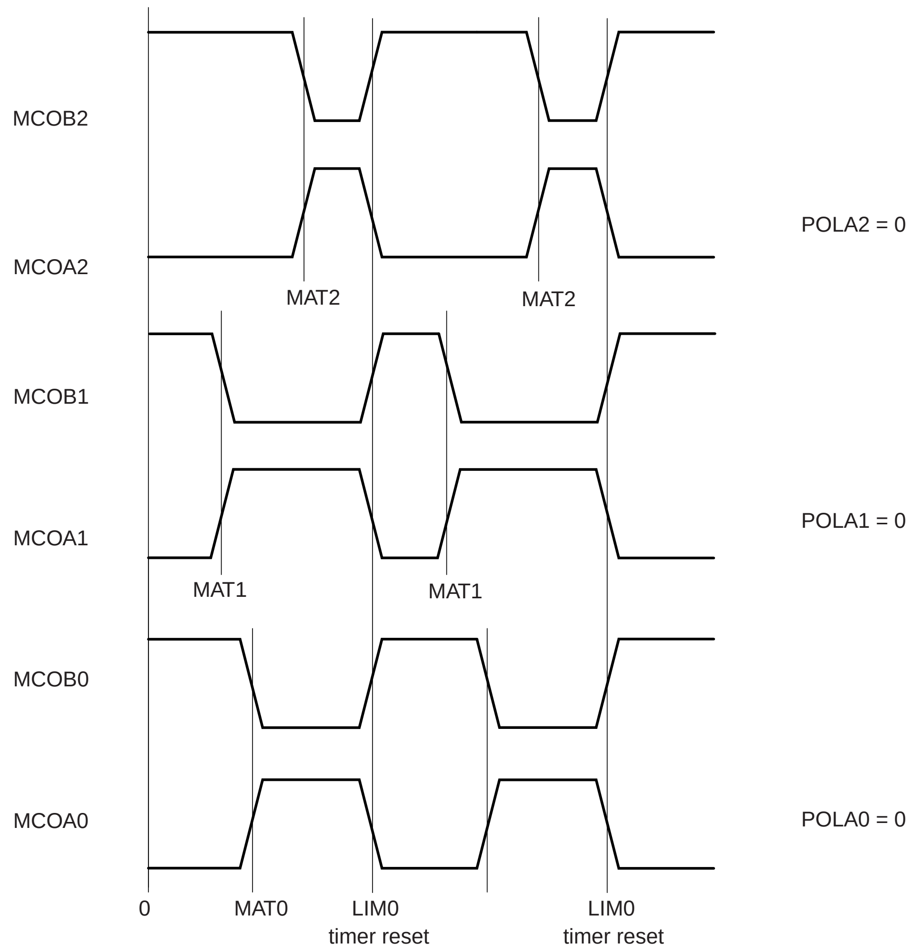

Fig 128. Three-phase AC mode sample waveforms, edge aligned PWM mode

25.8.8 Interrupts

The MCPWM includes 10 possible interrupt sources:

- When any channel's TC matches its Match register.
- When any channel's TC matches its Limit register.
- When any channel captures the value of its TC into its Capture register, because a selected edge occurs on any of MCI0-2.
- When all three channels' outputs are forced to "A passive" state because the MCABORT pin goes low.

Section 25.7.3 "MCPWM Interrupt registers" explains how to enable these interrupts, and Section 25.7.2 "MCPWM Capture Control register" describes how to map edges on the MCI0-2 inputs to "capture events" on the three channels.